Does the model fit the landscape? Shape and scale in Bayesian optimisation and CMA-ES

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Repository: https://github.com/isaacsoong055/DSA4212

Abstract

Bayesian optimisation (BO) and CMA-ES both carry an implicit model of the landscape: CMA-ES a Gaussian search distribution with a covariance (its shape) and a step size (its scale), BO a Gaussian-process surrogate whose kernel metric and lengthscales play the same two roles. We test one explanation in two settings: a method succeeds when its model's shape and scale can match the problem, whatever its family. *Part A (multimodality)* asks whether the model's scale lets a method *find* and then *refine* a narrow distant basin: neither default method finds it; CMA-ES's step size cannot do both jobs at once, and BO succeeds once its lengthscale is capped (1/50 to 44/50 on fresh seeds) while acquisition tuning helps far less. *Part B (geometry)* asks whether the model can *tilt*: on rotated, ill-conditioned valleys diagonal CMA-ES, the ARD kernel and TuRBO's axis-aligned trust region all fail, while their tilting counterparts do not, once BO's own fitting failures are removed. In both parts the decisive BO changes were to the model fit, not the acquisition function. All results are in 2-D and pre-registered predictions are reported whether they held or failed.

1 Introduction

1.1 Motivation and the two methods

Many optimisation problems give only function values: tuning a model's hyperparameters, calibrating a simulator, choosing a design. There are no gradients, and each evaluation can cost a training run or a simulation, so a method must decide where to evaluate next from the values it has seen, and its budget is counted in evaluations. Bayesian optimisation (BO) and the covariance matrix adaptation evolution strategy (CMA-ES) are two of the most widely used methods for this setting, and they embody opposite design choices: BO models the function, while CMA-ES adapts a search distribution.

BO [5] fits a Gaussian-process (GP) surrogate to all evaluations so far. Its kernel encodes smoothness and lengthscales, and its posterior gives a prediction and an uncertainty everywhere. An acquisition function, such as expected improvement (EI), turns these into the next point, trading off exploiting good predictions against exploring uncertain regions. *CMA-ES* [7] samples a population of λ candidates from a Gaussian $\mathcal{N}(m, \sigma^2 C)$, ranks them, moves the mean towards the best, and adapts the covariance C and the step size σ from successful steps. It uses only the ranking of the values. The basic trade-off follows: BO is sample-efficient because it uses every value through a global model, but each step is expensive (fitting the GP grows cubically with the number of points) and its success depends on that model being right; CMA-ES needs more evaluations but each step is nearly free, and it learns the *local* geometry of the problem.

1.2 What the literature says, and what it leaves open

Our questions came from the readings. Hansen's tutorial [7] states the invariances CMA-ES is designed to have: to any order-preserving transformation of f, and to rotations and other linear maps of the search space provided the initial conditions are mapped too; its covariance comes to approximate the inverse Hessian on a convex quadratic. This raised the question of which invariances survive in practice and whether a model that cannot rotate pays for it. Restart schemes that grow or mix the population size (IPOP, BIPOP [2, 6]) were introduced for multimodal functions, which raised the question of whether population and restarts help find a narrow basin. TuRBO [4] restricts BO's search to local trust regions whose sides are set per coordinate from the ARD lengthscales, an axis-aligned shape; COCO [9] sets the benchmarking practice we follow (budgets in evaluations, random instances, targets and intervals).

The literature also shows that BO's results depend on settings that are rarely reported. Hvarfner et al. [10] argue that BO's poor high-dimensional performance owes much to its lengthscale prior, and that a dimension-scaled prior makes plain BO competitive. Ament et al. [1] show that EI's values and gradients vanish numerically, so the acquisition optimiser can fail; and Siivola et al. [11] show that GP-based BO over-explores the boundary of the search box. A comparison of "BO against CMA-ES" run with library defaults therefore mixes the family with hidden choices: kernel fitting, lengthscale bounds, noise handling, step size. What remains open is which of those properties actually decides success.

1.3 Problem formulation

Research question. When BO and CMA-ES succeed or fail on a black-box problem, which property of the method decides the outcome, and does it matter more than the choice of family?

Guiding hypothesis. Each method carries an implicit model of the landscape. For CMA-ES it is the search distribution: the covariance sets its *shape* (orientation and relative stretch) and the step size its *scale*. For BO it is the GP: the kernel metric sets the shape (a full metric, or the diagonal ARD metric) and the lengthscales the scale. A method succeeds when its model can match the landscape's shape (orientation, conditioning) and scale (the size of the features it must find and resolve), and fails when a restriction or a fitting failure prevents it, whichever family it belongs to.

1.4 Shared experimental approach

Both parts fix a budget counted in objective evaluations, with initial designs and restarts included; run every variant on the same seeds, so comparisons are paired; change one setting at a time against a baseline; and commit their predictions to the repository before the runs that test them, reporting each as held, partly held or failed. A method's recommendation is its best evaluated point, judged on the true objective: in Part A by whether its regret reaches a success threshold, in Part B by its regret in decades of $\log_{10} f$. With noisy evaluations the recommended point is scored on the noise-free objective, and several recommendation rules are compared. Uncertainty is given as 95% intervals (Wilson and Newcombe intervals for success rates; paired bootstrap intervals over instances for regret). The two parts use different budgets, measures and BO implementations (BoTorch [3]; NumPy/SciPy), so their numbers are never put on common axes.

1.5 The two directions

Part A, multimodality (Isaac Soong), asks whether each method can *find* a narrow, distant global basin and then *refine* it, within 300 evaluations. It varies CMA-ES's population size, restarts and step size, and BO's acquisition function, GP fitting settings and kernel, on a decoy bowl with a narrow Gaussian well (whose width and distance are varied), with a sphere as a unimodal control and Rastrigin and Schwefel as validation functions.

Part B, geometry and parameterisation (Paul Ng), asks whether a method's model can *tilt* to a rotated, ill-conditioned valley, and which theoretical invariances survive the implementation. It varies the conditioning and rotation of the problem and whether each method's model can rotate (full against diagonal covariance; full against ARD kernel metric; a rotated against an axis-aligned trust region), and also tests changes of units, monotone transformations of f, the box boundary and noisy evaluations, on ellipsoids, Rosenbrock (COCO f8/f9 [8]) and a noisy quadratic, within 200 evaluations.

1.6 Preview of findings and contributions

Part A finds that neither default method discovers a narrow, distant basin. For CMA-ES no population size, restart scheme or step size fixes this, because one step size cannot both find the basin and refine it; for BO, capping the GP lengthscale does, while tuning the acquisition function helps far less. Part B finds that, in both families, the version of the method that cannot tilt fails under rotation and the version that can does not; for BO this became visible only after a fitting failure that stalled every run was diagnosed and fixed, and it then held across two kernels, two acquisition searches and a TuRBO-style trust region. CMA-ES keeps its rank invariance exactly; BO keeps neither invariance.

Together the parts contribute a tested account of shape and scale across both families; the shared, unplanned finding that BO's decisive fixes were to the *model fit*, not to the acquisition function; and a set of negative and surprising results reported as such: pre-registered predictions that failed, an initial negative result that turned out to be a fitting failure, a boundary trap, and BO runs that do not repeat bit for bit.

Roadmap. Section 2 presents Part A and Section 3 Part B, each opening with its answer, then setting out the evidence experiment by experiment, and closing with its predictions against outcomes. Section 4 draws the two together and states the limitations.

2 Part A: finding and refining a narrow basin (multimodality)

Answer in brief. Neither default method finds a narrow, distant well within 300 evaluations. For CMA-ES no setting we tried fixes this, because one step size cannot both find the well and refine it; for BO, capping the GP lengthscale does, while changing the acquisition function helps far less.

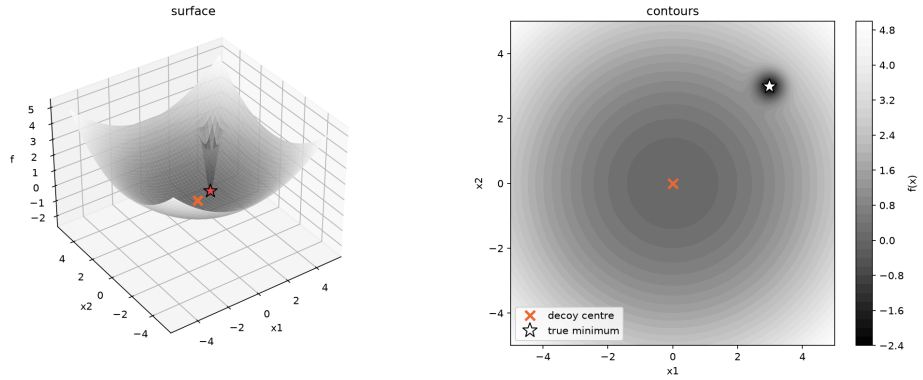


Figure 1. Part A. Primary landscape: a decoy bowl centred at the origin and a narrow global well at $(3, 3)$, $w = 0.3$.

2.1 Question, set-up and predictions

At a fixed budget of $N = 300$ evaluations, how do population size, restarts, step size and acquisition and surrogate choices change the probability of finding a narrow, distant global basin, and why? Predictions H1–H6 and M1–M3 were committed before any optimiser result was inspected; BO1–BO13 and CMA1–CMA6 were added later, each before the experiment that tested it, except BO1 (post hoc, labelled as such).

The primary landscape is a broad decoy bowl with a narrow, deep Gaussian well, $f(x) = \|x-c\|^2/s - A\exp(-\|x-x_g\|^2/2w^2)$ with $c = 0$, $s = 10$, $A = 4$, width $w = 0.3$, the well at $(3, 3)$ (Fig. 1), on $[-5, 5]^2$. A run succeeds when its regret falls to $\varepsilon = 0.05$, a threshold chosen from random sampling alone. CMA-ES is the `cma` library (default $\lambda = 6$, $\sigma_0 = 2$, start $x_0 \sim U[-4, 4]^2$); BO is BoTorch GP-EI (Matérn-5/2, 10 initial points, refit every 5 evaluations, lengthscale upper bound 20, fixed noise 10^{-6}). Each variant changes one setting and uses seeds 1000–1049; success rates carry Wilson intervals and differences Newcombe intervals.

2.2 Both defaults fail at discovery, not refinement (Exp 0)

Neither default method finds the well, and both fail because they never sample it, not because they cannot refine it. *Prediction.* H1: default CMA-ES gets stuck in the decoy basin. *Held. Result.* CMA-ES 0/50 $[0, 0.07]$, BO 5/50 $[0.04, 0.21]$, random sampling 2/50; the intervals overlap pairwise. *Why (measured).* CMA-ES's σ falls below 0.2 by generation 19 (median), and from evaluation 100 on at least 97.5% of its samples lie within 0.5 of the decoy centre. BO's lengthscale reaches its upper bound in 42 of 50 runs, none of which succeeds; in the 45 failures no sample ever enters the well, and every run that sampled it succeeded. *Caveat.* "early" collapse is read from the traces, not a pre-registered threshold.

2.3 Population and restarts do not help CMA-ES: one step size must both find and refine (Exps 1–3)

No population size or restart scheme lets CMA-ES succeed, and its step size explains why. *Predictions.* H2, a larger λ raises success: *failed.* H3, restarts beat a larger λ: *failed.* CMA1, from near the well success falls as σ_0 grows: *held. Result.* Success is 0/50 at $\lambda = 6, 12, 24$, also with five times the budget, although the number of runs that touch the well rises from 1 to 7 to 13. IPOP and BIPOP [2, 6] are identical at this budget, because only one restart fits, and give 0/50. From a start 1.0 from the well, $\sigma_0 = 0.3$ succeeds in 33/50 runs and the default $\sigma_0 = 2$ in 2/50 ($+0.62$ $[+0.45, +0.74]$; Fig. 2); from a random start no σ_0 exceeds 1/50. *Why (measured).* mean traces show that with $\sigma_0 = 2$ a single well sample is outvoted by the other selected points on the bowl, which drag the mean back to the decoy (10 of the 11 runs that touched the well end nearer the decoy). Finding the well needs a large σ and refining it a small one. *Caveat.* restarts are tested only once per run here, and success is not a function of σ_0/w alone: wide wells prefer a large σ_0 (CMA5 failed).

2.4 BO is held back by its model fit, not its acquisition function (Exps 4–7)

BO succeeds once its GP's lengthscale is capped; changing the acquisition function helps far less. *Predictions.* H4, BO finds the well only if it explores enough: *partly held.* BO7, a lengthscale cap does not help: *failed. Result.* EI with $\xi = 0.1$ reaches 17/50 ($+0.24$ $[+0.08, +0.39]$ over $\xi = 0$); a larger ξ touches the well more often but refines it less (17/45 down to 3/42 given a touch); LCB, Thompson sampling and BoTorch's default MAP recipe stay at 3–4/50. Capping the lengthscale at 1 gives 39/50 ($+0.68$ $[+0.51, +0.79]$; Fig. 3), and a cap of 0.25 falls back to 6/50. At cap 1, RBF matches Matérn-5/2 (39/50) but Matérn-3/2 and 1/2 collapse (6 and 3/50). *Why*

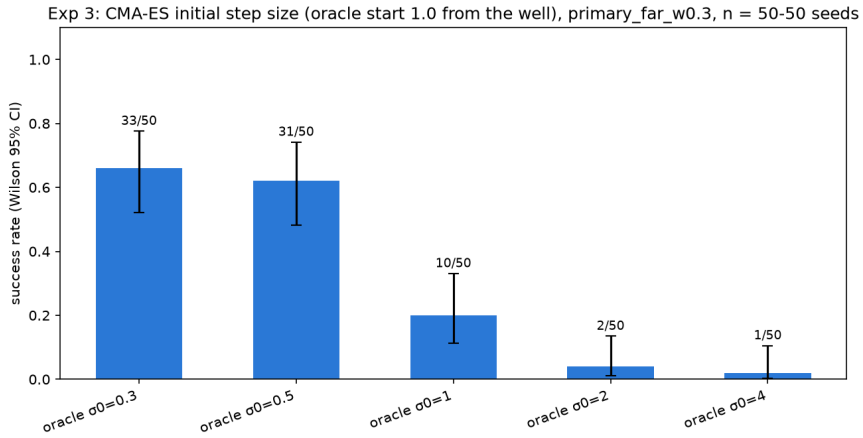


Figure 2. Part A, Exp 3. CMA-ES success against initial step size from a start 1.0 from the well (a diagnostic of refinement, not a method comparison).

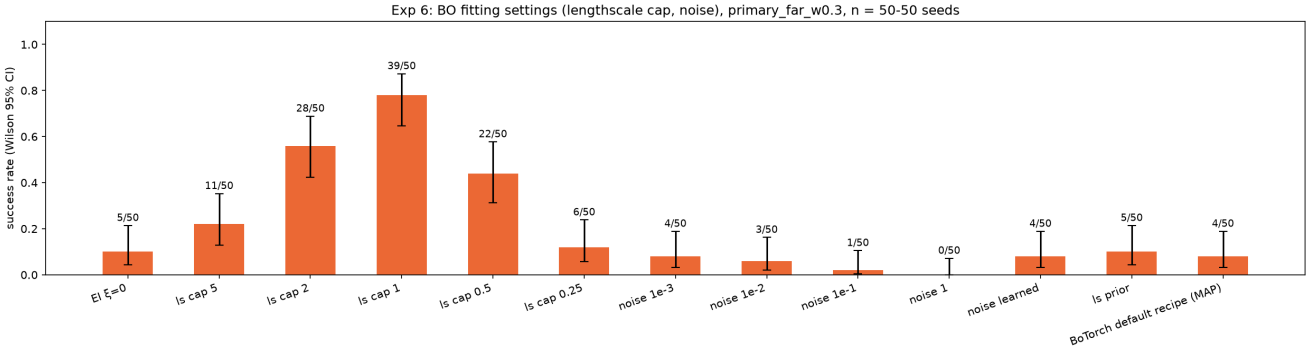


Figure 3. Part A, Exp 6. BO fitting settings changed one at a time: lengthscale cap and noise level (BoTorch, $N = 300$).

(measured). the default fit drives the lengthscale to its bound and the search stays near the bowl centre, visiting 10 of 100 grid cells after the initial design; with the cap it sweeps the box (about 100 of 100). The rough kernels fail at discovery: every run of theirs that touched the well succeeded. *Hypothesis.* the same cap means a different smoothness for each kernel, so rough kernels would need a tighter one (untested). *Caveat.* the cap was the best of five, so 39/50 is optimistic; the fresh seeds in Section 2.5 test that.

2.5 The capped BO holds across widths, costs on a unimodal bowl and transfers only partly (Exps 8–10)

On fresh seeds the cap's gain holds at every width from 0.3 up, but it has a cost on an easy bowl and transfers only partly to standard multimodal functions. *Predictions.* H5, a larger λ does not help on the sphere: *held.* H6, λ helps more on Rastrigin than on Schwefel: *not resolved. Result.* On seeds 2000–2049, not used to choose any setting, the capped BO succeeds in 44/50 at $w = 0.3$ and 50/50 at $w = 0.6$ and 1.0, against 1, 10 and 39/50 for default BO (Fig. 4). CMA-ES needs a wide well (35/50 at $w = 1.0$, 0/50 at $w = 0.3$), and its best variant ($\sigma_0 = 1$) gains nothing; on a near well default CMA-ES beats default BO (30 against 13/50 at $w = 0.3$). On the sphere the cap slows BO early (regret 8.9×10^{-4} against 1.2×10^{-6} at evaluation 100) and a larger λ worsens CMA-ES. On Rastrigin and Schwefel the cap's gain is small and uncertain, and BO is far stronger than CMA-ES on Schwefel (16/20 default, 20/20 capped, against at most 8/50). *Why.* not established; the raw runs were checked and no bug or configuration slip was found. *Caveat.* the two seed sets differ more than expected for some defaults (BO at $w = 0.6$: 23/50 against 10/50), unexplained; BO's validation runs use 20 seeds.

2.6 Predictions against outcomes

3 Part B: can the search tilt? (geometry and parameterisation)

Answer in brief. In both families, the version of the method that cannot tilt fails under rotation and the version that can does not; for BO this became visible only after a fitting failure was found and fixed. CMA-ES keeps its rank invariance exactly; BO keeps neither invariance, and its behaviour at the box edge and under noise depends

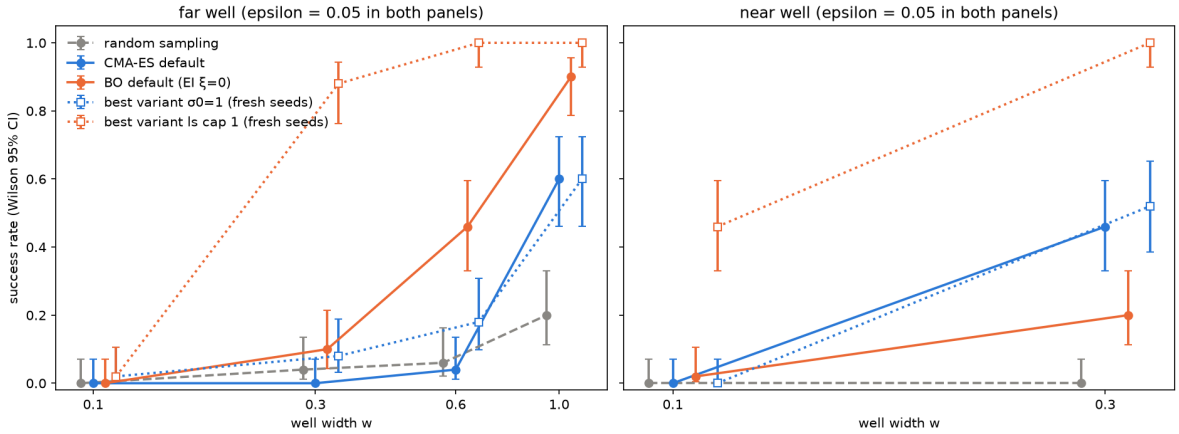


Figure 4. Part A, Exp 8. Success against well width for the defaults and the best variants (fresh seeds), far and near well.

Prediction (pre-registered)	Exp	Outcome	Evidence
H1: default CMA-ES gets stuck in the decoy basin	0	Held	0/50; $\sigma < 0.2$ by generation 19
H2: a larger λ raises success	1	Failed	0/50 at $\lambda = 6, 12, 24$, also at $N = 1500$
H3: restarts beat a larger λ	2	Failed	IPOP = BIPOP = 0/50
CMA1: near the well, success falls as σ_0 grows	3	Held	33/50 at $\sigma_0 = 0.3$, 2/50 at 2
CMA5: success is set by σ_0/w alone	3	Failed	wide wells prefer a large σ_0
H4: BO finds the well only if it explores enough	4–5	Partly held	$\xi = 0.1$: 17/50, falls beyond; flat in β
BO7: a lengthscale cap does not help	6	Failed	cap 1: 39/50 against 5/50
H5: a larger λ does not help on the sphere	9	Held	regret worse with λ
H6: λ helps more on Rastrigin than Schwefel	10	Not resolved	intervals overlap

Table 1. Part A. Selected pre-registered predictions and how they fared (full list in `multimodality/docs/`).

on implementation details.

3.1 Question, set-up and predictions

Near a minimum $f \approx f^* + \frac{1}{2} h^\top H h$. The Hessian's eigenvalues are the curvatures; their ratio κ is the conditioning (a narrow valley when large); its eigenvectors are the valley's orientation, which a rotation changes without changing κ. A full covariance (CMA-ES) or a full kernel metric $M = LL^\top$ (BO) can align with a rotated valley; a diagonal covariance, the standard ARD kernel and TuRBO's axis-aligned trust region [4] cannot. We predicted that (i) each axis-restricted method pays a penalty under rotation that its tilting counterpart does not, (ii) only when the problem is both ill-conditioned and rotated, and (iii) the tilting version pays a small price when no tilt is needed.

2-D ellipsoids with log-spaced eigenvalues from 1 to κ, the optimum on a radius-2 sphere, aligned or Haar-rotated, 15 instances (COCO-style; seed = instance). CMA-ES follows Hansen's tutorial, with a diagonal ablation. BO is GP-EI (initial design of $2d + 4$, EI over 200 uniform candidates polished by L-BFGS-B, targets log-transformed) with a MAP lengthscale prior on the metric's eigen-lengthscales [10], so that the prior does not favour axis-aligned metrics. Both get the box $[-5, 5]^2$ (BO searches it; CMA-ES starts at its centre with $\sigma_0 = 3$) and 200 evaluations. We report regret after that fixed budget because BO cannot run much longer (each run takes minutes), and give the runtime to a target, the measure COCO recommends [9], where the two views differ.

3.2 CMA-ES fails under rotation only when it cannot tilt (E1, E7)

Diagonal CMA-ES fails on a rotated, ill-conditioned valley and full CMA-ES does not. *Predictions.* (i)–(iii) for CMA-ES: *held. Result.* at $\kappa = 1{,}000$ diagonal CMA-ES solves every instance at 0° and 90° and none between 30° and 60°, while full CMA-ES solves every angle (Fig. 5); at $\kappa = 100$ both succeed everywhere. Full CMA-ES needs 405 against 408 median evaluations under Haar rotation, and up to 21.6% more evaluations than diagonal on aligned problems at $d = 32$. *Why (measured).* the 90° control only swaps the axes and recovers fully, so the rotation, not the conditioning, causes the failure; IPOP-style restarts leave it in place (7% against 20% reaching 10^{-5} at rotated $\kappa = 10^4$), because a restarted diagonal run still cannot tilt. *Caveat.* E1 runs to 4,000 evaluations and a target of 10^{-8}, longer than the 200-evaluation head-to-head.

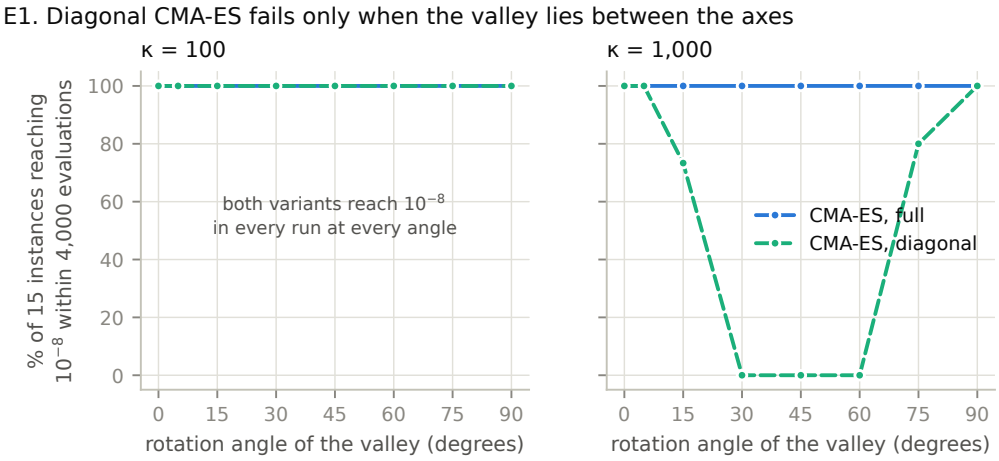


Figure 5. Part B, E1. Success of full (solid) and diagonal (dashed) CMA-ES against the angle of a 2-D valley.

BO variant (2-D, 15 instances, 200 evaluations)	rotated $\kappa = 10^4$	aligned $\kappa = 10^4$
EI, squared exponential, noise fixed	$+1.40$ [$+0.68, +1.69$]	-0.13 [$-1.15, +0.19$]
EI, Matérn-5/2	$+1.06$ [$+0.59, +1.43$]	-0.13 [$-0.70, +0.54$]
EI + local candidates	$+1.28$ [$+0.12, +2.36$]	-2.96 [$-3.44, +2.61$]
Trust region, ARD box vs rotated box	$+2.92$ [$+1.27, +3.79$]	-0.39 [$-3.11, +4.65$]

Table 2. Part B. Axis-restricted minus tilting version, median $\log_{10} f$ at 200 evaluations, paired by instance (+ = axis-restricted worse). Wide aligned intervals come from the boundary trap.

3.3 BO first showed no penalty, because a fitting failure stalled every run (E2, E2e)

BO's first head-to-head contradicted the explanation, and the cause was a fitting failure, not the geometry. *Prediction.* (i) for BO, that ARD pays under rotation: *failed* on the first run. *Result.* ARD's learned long axis is off by exactly 45° on 45° problems (the full metric is within 0.2°–4°), so ARD cannot tilt; yet its rotation penalty was $+0.05$ [$-0.53, +0.63$] decades at $\kappa = 100$, while diagonal CMA-ES paid $+1.70$ [$+0.41, +2.57$]. BO stalled near $10^{-2.6}$ even on the round bowl. *Why (measured).* per-iteration logs ruled out the target transform's floor and the candidate search, and showed the GP fitting a noise term (SD 0.2–0.4) to a noise-free function: it explained its best points away as noise, its prediction at the incumbent sat 2–2.5 times the gap to the second-best point above the observed value, and EI underflowed below 10^{-12} in 97–100% of late iterations. Fixing the noise variance at 10^{-6} lifted BO by -1.17 [$-1.71, -0.74$] decades; LogEI [1], which fixes the underflow but not its cause, changed nothing ($+0.01$ [$-0.79, +0.47$]). *Caveat.* the fix assumes a noise-free objective; Section 3.7 tests it under noise.

3.4 Once BO works, the axis-restricted version pays in every variant (E2, E8, E10)

With the fix, BO-ARD pays a rotation penalty and BO-full does not, across two kernels, two acquisition searches and a trust region. *Predictions.* (i) and (ii) for BO: *held*. *Result.* paired by instance on one machine, BO-ARD's penalty became $+2.01$ [$+1.28, +2.97$] decades at $\kappa = 10^4$ and BO-full's stayed non-significant (Fig. 6); Table 2 repeats the comparison with a Matérn-5/2 kernel, 200 extra candidates around the incumbent, and a TuRBO-style trust region [4] whose box is axis-aligned with ARD-scaled sides or rotated to the full metric. Measured by runtime to a target, the picture of BO changes with its search: on the round bowl CMA-ES reaches 10^{-5} in every run (median 142–154 evaluations), BO with the noise fix in only 0–27% of runs within 200, and BO with local candidates in every run (median 41–56). Across families, the trust region beats full CMA-ES at 200 evaluations on aligned $\kappa = 100$ (-1.36 [$-2.02, -0.85$]) and rotated $\kappa = 10^4$ (-3.32 [$-4.24, -1.82$]); with plain EI, CMA-ES wins by 200 evaluations at $\kappa \leq 100$. *Why (measured).* the ARD kernel and the published trust region both scale each coordinate separately, so neither can align with a rotated valley. *Caveat.* long BO runs are not bit-repeatable, so the results hold in distribution; the wide aligned intervals in Table 2 come from the boundary trap (Section 3.6).

3.5 On a bending valley, local orientation and local search decide (E5, E9, E12)

On Rosenbrock, whose valley bends, what matters is the local orientation, and for BO local *search* helps while a locally *fitted* model does not. *Predictions.* that both CMA-ES variants converge with a long budget: *failed*. That a GP fitted only to nearby points follows the bend better: *failed*. *Result.* With 4,000 evaluations full CMA-ES reaches 10^{-5} in every run (median about 500 evaluations) and diagonal in 0% (f8) and 27% (f9) of runs. For BO

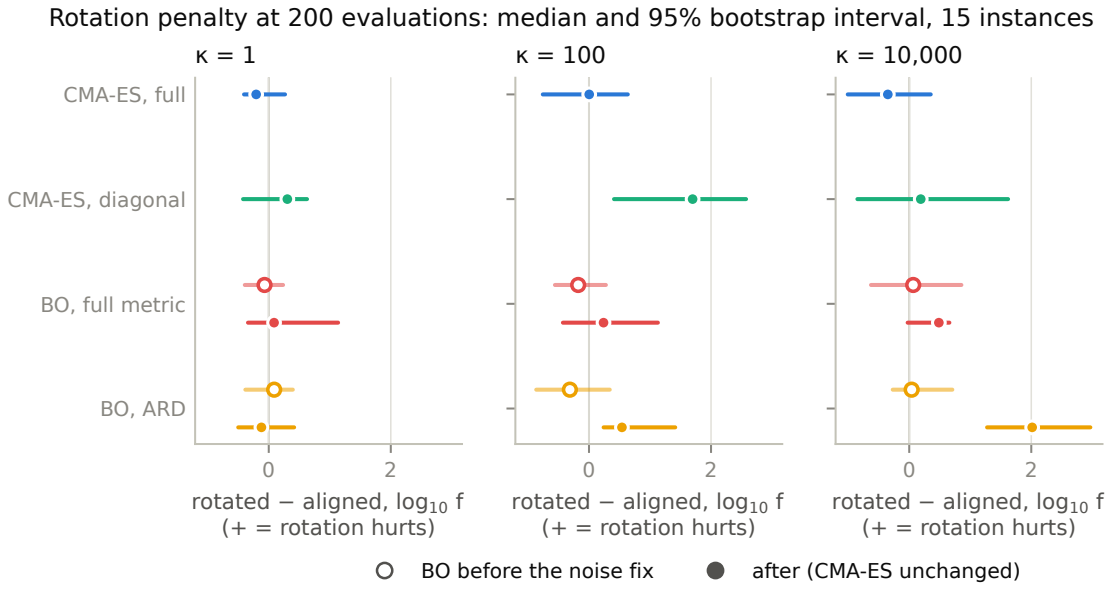


Figure 6. Part B. Rotation penalty at 200 evaluations (median, 95% bootstrap interval, 15 instances). Hollow: BO before the noise fix. The axis-restricted variant pays in both families once BO is no longer stalled.

at 300 evaluations, the trust region beats plain BO by 3.5–3.7 decades and its tilted box beats the axis-aligned one by 2.0–2.5 decades, even on f8; a trust region whose GP is fitted only to the points inside it (at least 20) trails the ordinary trust region by 1.8–2.0 decades, though it stays about 1.7 ahead of plain BO. *Why (measured).* even "aligned" Rosenbrock is tilted 63.5° near its optimum (local $\kappa \approx 2{,}500$), so no fixed axis alignment fits it. Plain BO's kernel adapts its lengthscales but is stationary, one metric for the whole box, which can only average the directions along a curving valley; inside a small box the valley is close to a straight, tilted one. On a quadratic control, where the shape is the same everywhere and a local fit can gain nothing, local fitting lost 2.7 decades, so its loss comes from fitting fewer points, not from geometry. TuRBO's authors found the opposite: local GPs predicted better than a global GP on a 14-D robot-pushing problem, with 200 points per region on an objective whose noise and lengthscales vary by region [4, Sec. 3.6]. *Hypothesis.* local fitting pays when each region holds enough data and the function really differs between regions, which neither our 2-D runs nor our smooth valley provide. *Caveat.* the trust region also scores 1,000 candidates inside its box against plain BO's 200 across the whole box, so part of its lead may come from that denser local search.

3.6 CMA-ES keeps its rank invariance; BO keeps neither, and the box edge can trap it (E3, E4, E11)

CMA-ES's invariances survive the implementation under the stated conditions; BO has none of them, and its interaction with the box edge depends on implementation details. *Predictions.* CMA-ES is exactly invariant to monotone maps of f: *held.* BO is nearly invariant to powers of f: *failed.* Reflecting candidates removes the boundary trap: *partly held. Result.* CMA-ES runs were bit-identical to the baseline under every monotone map of f (120/120); cubing f cost BO 1.7 decades. A change of units hurts every method with default set-ups (CMA-ES about 3.9 decades, BO 1.3–1.4), and mapping CMA-ES's initial covariance as Hansen requires ($C_0 = S^{-2}$) restores invariance. BO is 1.62 / 0.73 decades worse when the optimum lies 0.1 inside a box edge and better when it lies on it; with search around the incumbent, 4–8 of 15 runs per aligned ill-conditioned cell stall on the valley floor at the edge. Reflecting out-of-box candidates instead of clipping them removes this for the trust region (6 and 4 stuck runs to 0) but not for local candidates. *Why (measured).* cubing moves BO's relative target floor up to $f \approx 14$; GP posterior variance is largest at the edge [11]; in trapped runs about two random candidates per iteration are truly better, yet the GP gives them EI near 10^{-34}, a model failure. *Caveat.* why the GP mis-rates those points is untested; BO's own units invariance (per-input normalisation) is argued, not run.

3.7 Under noise, the recommendation rule matters as much as the budget split (E5, E6, E6b)

With noisy evaluations, how the final point is chosen matters as much as how the budget is spent, and repeats do not pay at this budget. *Predictions.* that fixing BO's noise backfires on noisy problems: *failed.* That BO's posterior-mean pick beats its lowest reading: *held. Result.* on the $\kappa = 100$ quadratic with additive noise $\sigma \in \{0.01, 0.1, 1\}$,

every recommendation scored on the noise-free f, fixed-noise BO still beats fitted-noise BO (-0.41 [-0.73, -0.10] at $\sigma = 0.01$); BO's posterior-mean pick beats its lowest reading by up to 1.08 decades; CMA-ES's mean helps only with $\lambda = 24$; repeats and a final re-evaluation reserve never helped significantly; with each family's best settings the two are level at every σ. *Why (measured).* when BO's model-based pick did worse (with averaged replicates), the noise-fitting GP had absorbed the best averages as noise; a correctly specified GP on the raw values never did worse than the lowest reading. *Caveat.* one problem, one noise model and one budget.

3.8 Predictions against outcomes

Prediction (written before the runs)	Exp	Outcome	Evidence
(i) diagonal CMA-ES pays under rotation	E1	Held	0% solved at 30°–60°, $\kappa = 1{,}000$
(i) BO-ARD pays under rotation (first run)	E2	Failed	+0.05 [-0.53, +0.63]: BO was stalling
(i) BO-ARD pays once BO works	E2, E8, E10	Held	+1.06 to +2.92 decades, rotated $\kappa = 10^4$
(ii) no penalty on aligned problems	E1, E2, E10	Held	aligned differences not significant
(iii) tilting costs a little when not needed	E1, E10	Held	+21.6% evaluations; ARD box 0.57 better
CMA-ES is exactly invariant to monotone maps of f	E3	Held	120/120 runs bit-identical
BO is nearly invariant to powers of f	E3	Failed	cube of f: +1.7 decades
fixing BO's noise backfires on noisy problems	E5	Failed	fixed noise still better, -0.41
both CMA-ES variants solve Rosenbrock with 4,000 evaluations	E9	Failed	diagonal: 0% (f8), 27% (f9)
a locally fitted trust region follows the bend better	E12	Failed	1.8–2.0 decades worse; control 2.7 worse

Table 3. Part B. Selected predictions and how they fared (all 21 in `geometry/`).

4 Synthesis: what the two parts establish together

Shape and scale both decide success, within each family. Part A shows that a model with the wrong *scale* fails to find or to refine a narrow basin, and Part B that a model restricted in *shape* fails on tilted valleys in both families. In neither part does the family alone predict the outcome: diagonal and full CMA-ES differ by up to 100 percentage points on the same rotated problem, and capped and default BO by 43 successes out of 50 on the same landscape (44 against 1 on fresh seeds).

BO's decisive fixes were to the model fit, not the acquisition. Independently, both parts found BO failing because its GP was mis-fitted, and both found acquisition changes nearly powerless: in Part A the lengthscale ran to its bound (capping it took success from 5 to 39/50; ξ, LCB and Thompson sampling at most 17/50), in Part B the fitted noise absorbed the best points (fixing it gained over a decade; LogEI nothing). In both, BO's results depend on implementation choices that published defaults leave open.

CMA-ES adapts shape well but must commit to a scale. One step size cannot both find a distant basin and refine a narrow one (Part A), but its covariance follows a rotated valley reliably (Part B); population size and restarts, which change only how much is sampled, did not move success on the primary landscape.

Trade-offs. BO leads early in both parts and costs roughly 10^5 times more computation per run; settings that help on the hard landscape cost on easy ones (the cap on the sphere, tilting on aligned problems); and BO's boundary preference appears in both parts (edge sampling under large ξ; the boundary trap).

Limitations. All headline results are 2-D, so dimension effects are inferred (for CMA-ES) rather than measured for BO. Each part rests on few landscapes (ellipsoids and Rosenbrock; one primary landscape plus two validation functions) and 15–50 seeds, which leaves some differences unresolved. BO results reproduce in distribution only: long BO runs are not bit-repeatable even with one BLAS thread. Some follow-ups were added after seeing results and are labelled as such; untested mechanisms are stated as hypotheses.

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